

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year: 2025

Programme: BCSIT

Full Marks: 100

Course: Mathematics II

Pass Marks: 45

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section "A"

Very Short Answer Questions

Attempt all the questions. [10×2=20]

1. Find the modulus of the complex number $z = 5 - 12i$.
2. Prove that: $\omega^3 = 1$
3. State the direct comparison test of convergence and divergence of infinite series.
4. Integrate: $\int_0^2 x e^x dx$
5. Find the partial derivative with respect to x and Partial derivative with respect to y of $f(x, y) = x^2y + 2xy + xy^2$.
6. What is the degree and order of the differential equation
$$\left[\frac{d^4y}{dx^4}\right]^2 + 2\left[\frac{d^3y}{dx^3}\right]^5 + 7\frac{dy}{dx} = -3.$$
7. What is the fundamental theorem of arithmetic? Give one example.
8. Find the period of $f(x) = \sin nx$.
9. Examine the whether the function $f(x) = 3x^3 - 4x$ is odd or even.
10. Solve the following differential equation by using variable separable method : $xdy - ydx = 0$

Section "B"

Descriptive Answer Questions

Attempt any five questions. [5×10=50]

11. a. Apply De-Moivre's theorem to find the value of $(1 + i)^{20}$
b. Find the square root of $-4 + 3i$.
12. a. Examine the convergence of the series $\sum_{n=1}^{\infty} \frac{n^2}{n^3-3}$ by using limit comparison test.
b. Test the convergence of the series by using D'Alembert's ratio test: $\frac{1}{3} + \frac{2}{3^2} + \frac{3}{3^3} + \frac{4}{3^4} + \dots$

13. a. Find the volume of the solid generated by revolving the region bounded by the curve $y = \sqrt{x}$ and the lines $y = 1$, $x = 4$ about the line $y = 1$.
- b. If $V = \ln(x^2 + y^2 + z^2)$ prove that $V_{xx} + V_{yy} + V_{zz} = \frac{2}{x^2 + y^2 + z^2}$
14. a. Find the optimum values of the function $f(x, y) = 300x + 270y - 10x^2 - 10y^2 - 10xy$.
- b. Integrate: $\int_0^1 3x^5 e^{x^3} dx$
15. Solve any two:
- a. $\frac{dy}{dx} + 2y = e^{-3x}$
- b. $(x^2 + y^2)dy = xydx$
- c. $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^{5x}$
16. a. Calculate GCD and LCM of 30 and 24 using division algorithm.
- b. Find the Fourier series for the function $f(x) = x^2$ in the interval $0 \leq x \leq 2\pi$.

Group "C"

Long Answer Questions

Attempt any two questions. $[2 \times 15 = 30]$

17. a. Find the real and imaginary part of the complex number $(\frac{\bar{W}}{Z})^2$ if $Z = 4 - 5i$ and $W = i - 6$. [7]
- b. Determine if the following series is convergent or divergent by using alternative series test. [8]
- $$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$
18. a. Define Gamma and beta function. Show that $\int_0^{\frac{\pi}{2}} \sin^4 x \cdot \cos^2 x dx = \frac{\pi}{32}$. [8]
- b. Find the general and particular solution for the differential equation: $\frac{d^2y}{dx^2} = 6x + 8$, $f'(5) = -18$, $f(2) = 30$. [7]
19. a. Express $f(x) = x$ as half range sine series in $0 < x < 2$ [7]
- b. Define prime. If p is prime and $p|ab$ then prove that $p|a$ or $p|b$. [8]